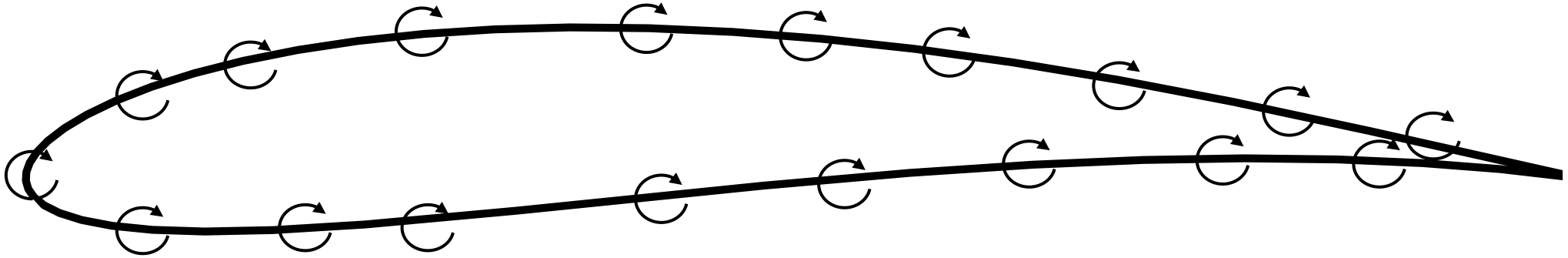
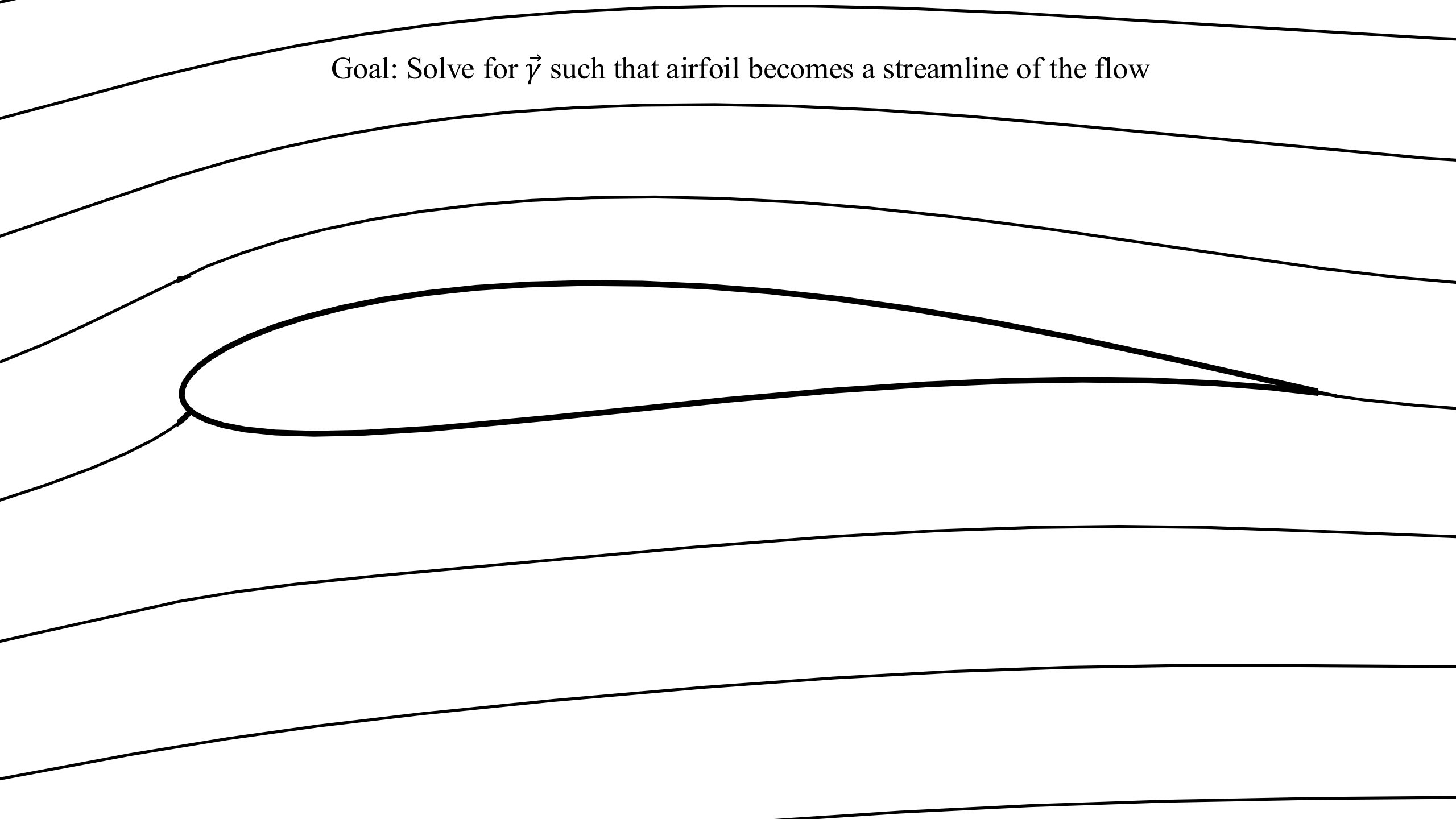
Vortex panel method

overview



Goal: Solve for $\vec{\gamma}$ such that airfoil becomes a streamline of the flow



Example (suggested code approach)

You'll be given a list containing the x,y coordinates of the nodal points defining your geometry. (in this example case, 12 points)

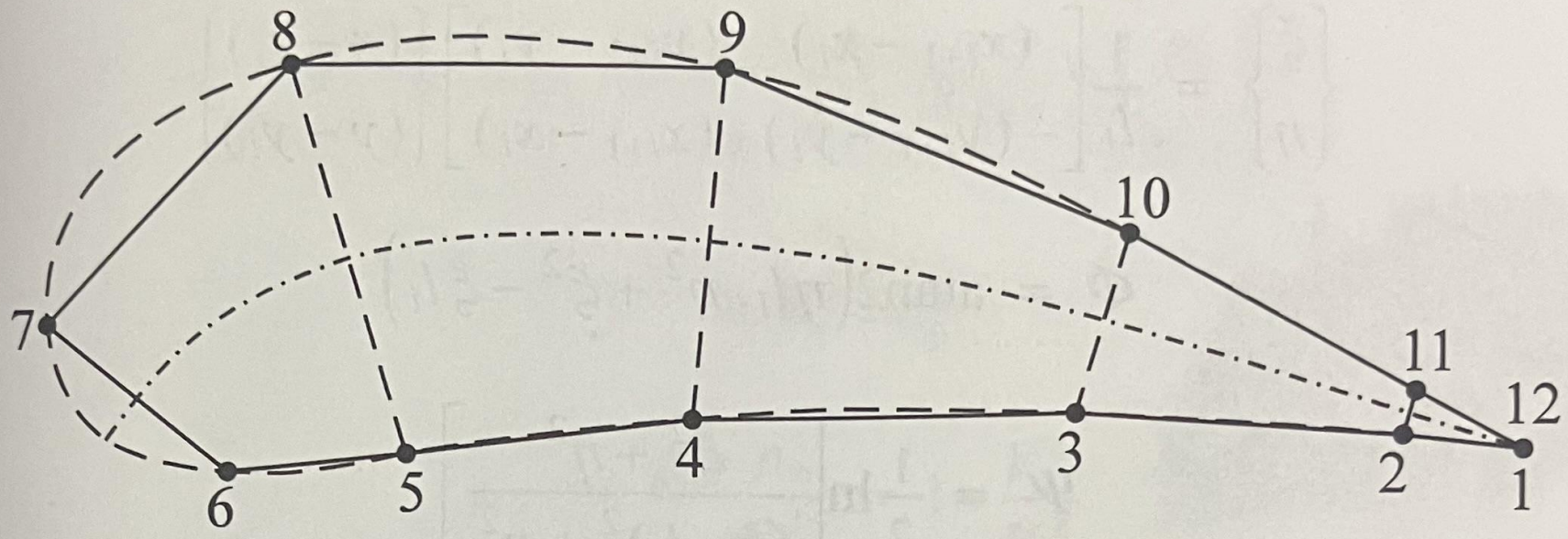


Figure 1.6.4. Vortex panel distribution with cosine clustering and an even number of nodes.

Step 1

Store the points in an $N \times 2$ array. (N is the number of rows- corresponds to the number of given points, and there are 2 columns for the x and y components of the nodal points)

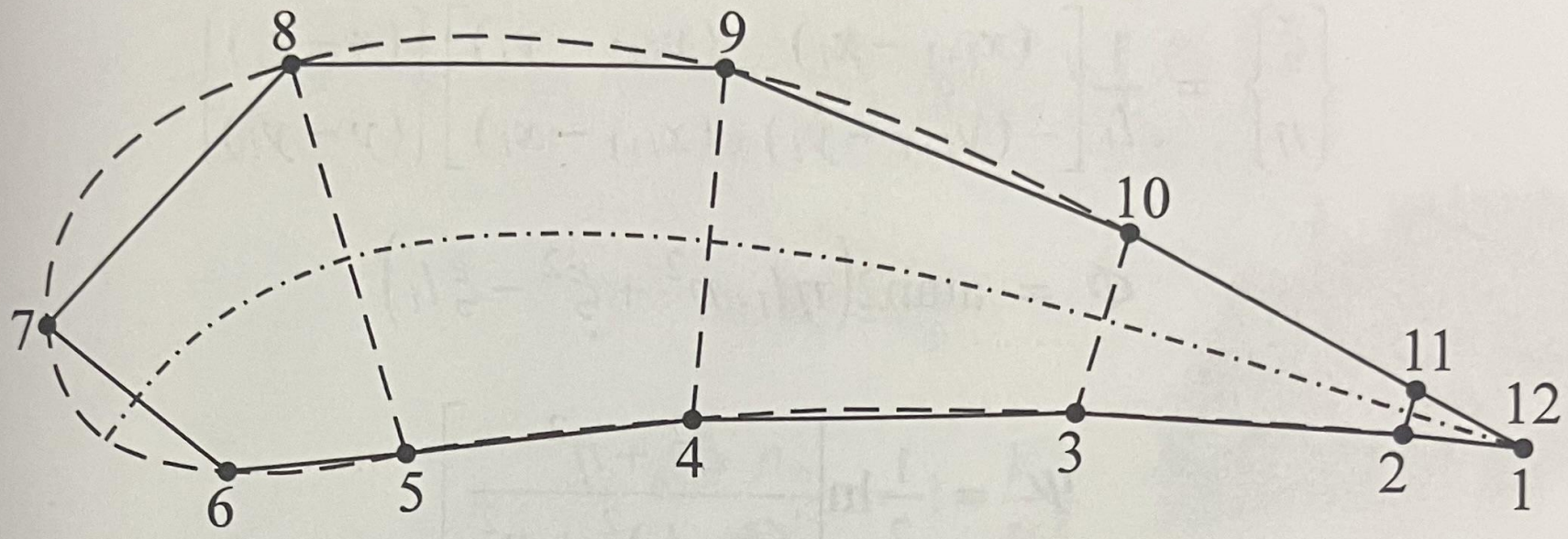


Figure 1.6.4. Vortex panel distribution with cosine clustering and an even number of nodes.

Step 2

Identify and store the number of panels: $N_{panels} = N - 1$ (where N is number of nodes)

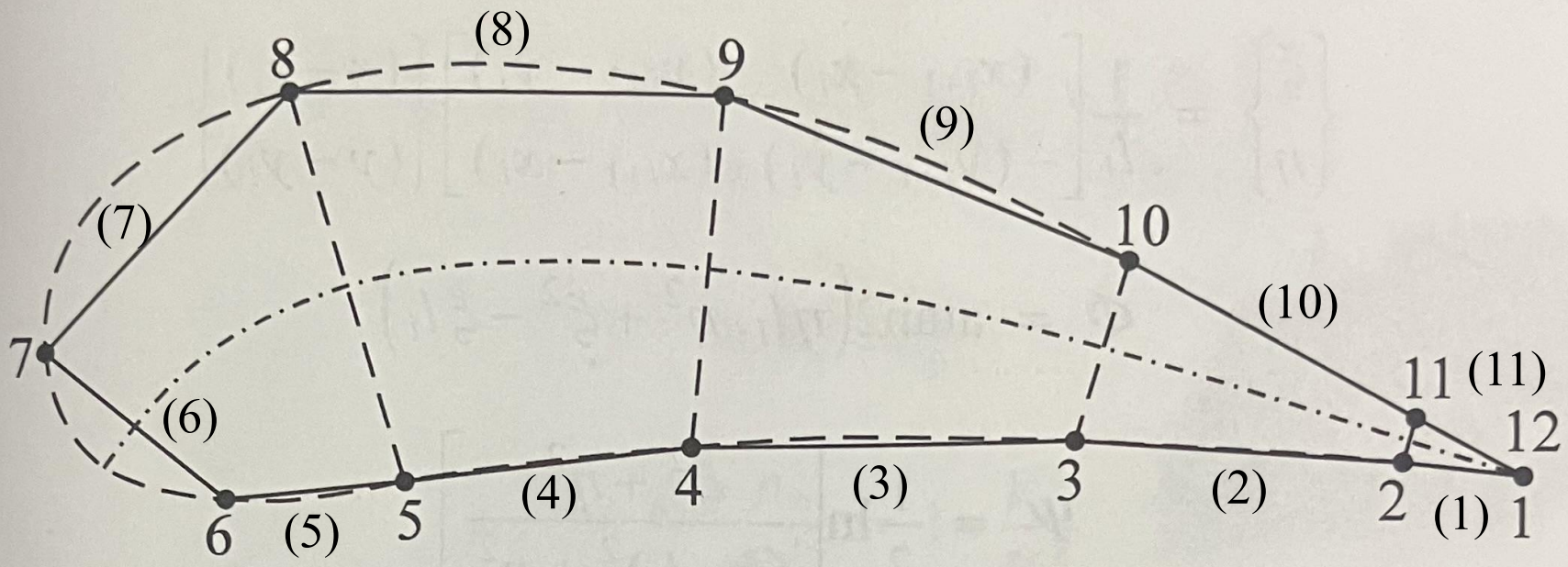


Figure 1.6.4. Vortex panel distribution with cosine clustering and an even number of nodes.

Step 3

Calculate lengths of each panel: $\ell_j = \sqrt{\left(x_{j+1} - x_j\right)^2 + \left(y_{j+1} - y_j\right)^2}$ and store in an $N_{panel} \times 1$ "lengths" array. There's only one column in this array; you're storing a single scalar value in each entry.

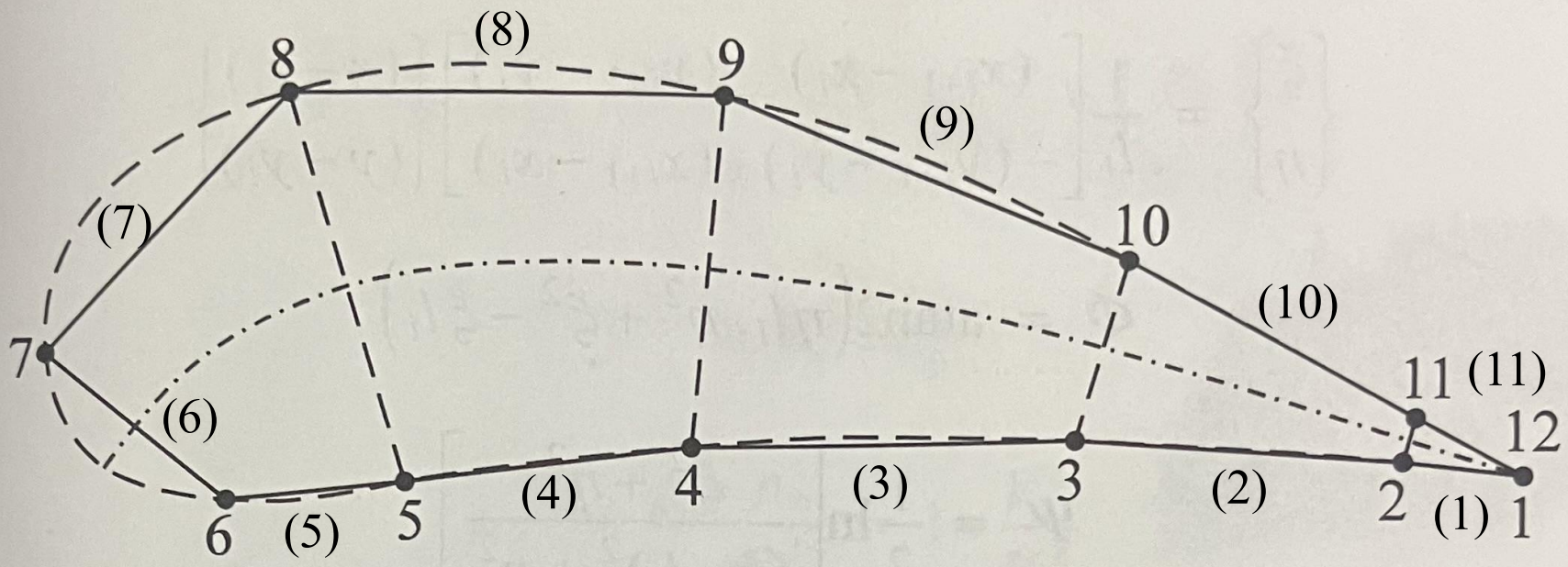


Figure 1.6.4. Vortex panel distribution with cosine clustering and an even number of nodes.

Step 4

Calculate and store the location of each control point: $\begin{Bmatrix} x_{cp}(j) \\ y_{cp}(j) \end{Bmatrix} = 0.5 \begin{Bmatrix} x_N(j) + x_N(j+1) \\ y_N(j) + y_N(j+1) \end{Bmatrix}, j = 1, N_{panels}$

and store in an $N_{panel} \times 2$ "control_points" array

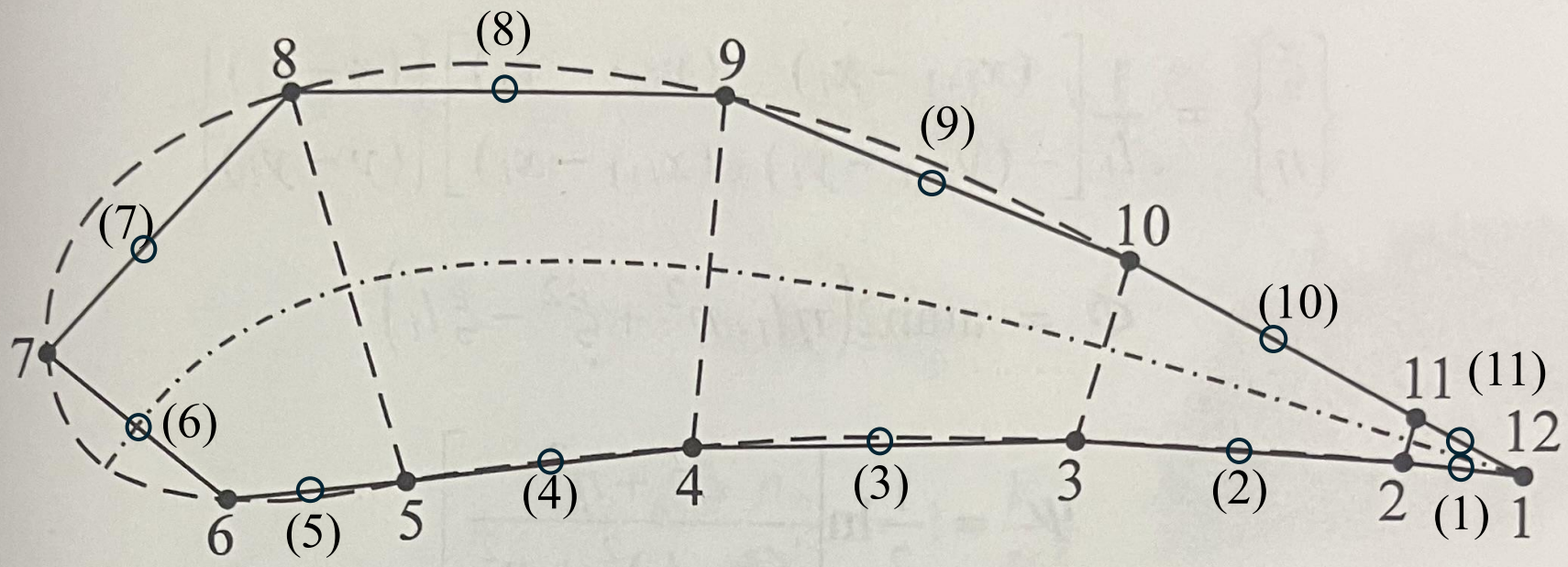


Figure 1.6.4. Vortex panel distribution with cosine clustering and an even number of nodes.

Step 5

Assemble the "A" matrix. Recall the following algorithm.

$$A_{ij} = 0.0, \quad i = 1,n; \quad j = 1,n \tag{1.6.24}$$

$$\left.\begin{aligned} A_{ij} &= A_{ij} + \frac{x_{i+1}-x_i}{l_i} P_{21_{ji}} - \frac{y_{i+1}-y_i}{l_i} P_{11_{ji}} \\ A_{ij+1} &= A_{ij+1} + \frac{x_{i+1}-x_i}{l_i} P_{22_{ji}} - \frac{y_{i+1}-y_i}{l_i} P_{12_{ji}} \end{aligned}\right\}, \quad i = 1,n-1; \quad j = 1,n-1 \tag{1.6.25}$$

$$A_{n1} = 1.0 \tag{1.6.26}$$

$$A_{nn} = 1.0 \tag{1.6.27}$$

where

$$[\mathbf{P}]_{j(x,y)} = \frac{1}{2\pi l_j^2}\begin{bmatrix} (x_{j+1}-x_j) & -(y_{j+1}-y_j) \\ (y_{j+1}-y_j) & (x_{j+1}-x_j) \end{bmatrix}\begin{bmatrix} [(l_j-\xi)\Phi+\eta\Psi] & (\xi\Phi-\eta\Psi) \\ [\eta\Phi-(l_j-\xi)\Psi-l_j] & (-\eta\Phi-\xi\Psi+l_j) \end{bmatrix} \tag{1.6.23}$$

$$\begin{Bmatrix} \xi \\ \eta \end{Bmatrix} = \frac{1}{l_j}\begin{bmatrix} (x_{j+1}-x_j) & (y_{j+1}-y_j) \\ -(y_{j+1}-y_j) & (x_{j+1}-x_j) \end{bmatrix}\begin{Bmatrix} (x-x_j) \\ (y-y_j) \end{Bmatrix} \tag{1.6.20}$$

$$\Phi = \operatorname{atan2}\left(\eta l_j, \eta^2+\xi^2-\xi l_j\right) \tag{1.6.21}$$

$$\Psi = \frac{1}{2}\ln\left[\frac{\xi^2+\eta^2}{(\xi-l_j)^2+\eta^2}\right] \tag{1.6.22}$$

Let's break this down into details

Step 5.1

$$A_{ij} = 0.0, \quad i = 1,n; \quad j = 1,n \tag{1.6.24}$$

Pre-allocate an $N \times N$ matrix and fill with zeros.

```
import numpy as np
A_matrix = np.zeros((N,N)) # assumes N has been defined (number of nodes)
```

$$A = \begin{bmatrix} 0 & - & 0 \\ | & \backslash & | \\ 0 & - & 0 \end{bmatrix} \ (N \times N \text{ matrix with zeros})$$

Step 5.2

Loop through the existing A-matrix and populate using (1.6.25). The rows are i and the columns are j.

$$\left.\begin{aligned} A_{ij} &= A_{ij} + \frac{x_{i+1}-x_i}{l_i} P_{21_{ji}} - \frac{y_{i+1}-y_i}{l_i} P_{11_{ji}} \\ A_{ij+1} &= A_{ij+1} + \frac{x_{i+1}-x_i}{l_i} P_{22_{ji}} - \frac{y_{i+1}-y_i}{l_i} P_{12_{ji}} \end{aligned}\right\}, \quad i = 1, n-1; \quad j = 1, n-1 \qquad (1.6.25)$$

where

$$[\mathbf{P}]_{j(x,y)} = \frac{1}{2\pi l_j^2} \begin{bmatrix} (x_{j+1}-x_j) & -(y_{j+1}-y_j) \\ (y_{j+1}-y_j) & (x_{j+1}-x_j) \end{bmatrix} \begin{bmatrix} [(l_j-\xi)\Phi + \eta\Psi] & (\xi\Phi - \eta\Psi) \\ [\eta\Phi - (l_j-\xi)\Psi - l_j] & (-\eta\Phi - \xi\Psi + l_j) \end{bmatrix} \qquad (1.6.23)$$

Let's talk about the P-matrix

Step 5.2 (continued)

There are $N-1$ $[\mathbf{P}]$ matrices which affect the velocity at each control point i

$$[\mathbf{P}]_{j(x,y)} = \frac{1}{2\pi l_j^2}\begin{bmatrix}(x_{j+1}-x_j) & -(y_{j+1}-y_j)\\ (y_{j+1}-y_j) & (x_{j+1}-x_j)\end{bmatrix}\begin{bmatrix}[(l_j-\xi)\Phi+\eta\Psi] & (\xi\Phi-\eta\Psi)\\ [\eta\Phi-(l_j-\xi)\Psi-l_j] & (-\eta\Phi-\xi\Psi+l_j)\end{bmatrix} \quad (1.6.23)$$

$[\mathbf{P}]_j(x,y)$ is the geometric influence of panel j on some point $p(x,y)$. When multiplied by $\begin{Bmatrix}\gamma_j\\ \gamma_{j+1}\end{Bmatrix}$, it yields the velocity induced by panel j on $p(x,y)$. **When solving our system, the point $\boldsymbol{p(x,y)}$ is the location of control point $\boldsymbol{i}$**

ξ and η then define the panel-coordinate location of control point i. These are calculated as follows:

Coordinates of control point i

$$\begin{Bmatrix}\xi\\ \eta\end{Bmatrix} = \frac{1}{l_j}\begin{bmatrix}(x_{j+1}-x_j) & (y_{j+1}-y_j)\\ -(y_{j+1}-y_j) & (x_{j+1}-x_j)\end{bmatrix}\begin{Bmatrix}(x-x_j)\\ (y-y_j)\end{Bmatrix} \quad (1.6.20)$$

$$\Phi = \operatorname{atan2}\left(\eta l_j, \eta^2+\xi^2-\xi l_j\right) \quad (1.6.21)$$

$$\Psi = \frac{1}{2}\ln\left[\frac{\xi^2+\eta^2}{(\xi-l_j)^2+\eta^2}\right] \quad (1.6.22)$$

After ξ and η have been computed, Φ and Ψ can be calculated, and $[\mathbf{P}]_j(x,y)$ can be solved!

Step 5.2 (continued)

After matrix multiplication, [**P**] will be a 2×2 matrix like this:

$$[\mathbf{P}] = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \rightarrow \textit{can be understood as} \rightarrow [\mathbf{P}] = \begin{bmatrix} \text{influence of } \gamma_1 \text{ on } V_x & \text{influence of } \gamma_2 \text{ on } V_x \\ \text{influence of } \gamma_1 \text{ on } V_y & \text{influence of } \gamma_2 \text{ on } V_y \end{bmatrix}$$

$$\left.\begin{aligned} A_{ij} &= A_{ij} + \frac{x_{i+1}-x_i}{l_i} P_{21_{ji}} - \frac{y_{i+1}-y_i}{l_i} P_{11_{ji}} \\ A_{ij+1} &= A_{ij+1} + \frac{x_{i+1}-x_i}{l_i} P_{22_{ji}} - \frac{y_{i+1}-y_i}{l_i} P_{12_{ji}} \end{aligned}\right\}, \quad i = 1, n-1; \quad j = 1, n-1 \quad (1.6.25)$$

What do these terms mean?

$$A_{ij} = A_{ij} + \frac{x_{i+1}-x_i}{l_i} P_{21_{ji}} - \frac{y_{i+1}-y_i}{l_i} P_{11_{ji}}$$

$$A_{ij+1} = A_{ij+1} + \frac{x_{i+1}-x_i}{l_i} P_{22_{ji}} - \frac{y_{i+1}-y_i}{l_i} P_{12_{ji}}$$

$$\hat{n_j} = \frac{1}{\ell_j}\begin{Bmatrix} -t_y \\ t_x \end{Bmatrix} = \begin{Bmatrix} -\dfrac{y_{j+1}-y_j}{\ell_j} \\ \dfrac{x_{j+1}-x_j}{\ell_j} \end{Bmatrix} = \begin{Bmatrix} n_x \\ n_y \end{Bmatrix}$$

$$A_{ij} = A_{ij} + \frac{x_{i+1}-x_i}{l_i} P_{21_{ji}} - \frac{y_{i+1}-y_i}{l_i} P_{11_{ji}}$$

This is the velocity influence contribution of node j of panel j dotted in the normal direction (added to the current normal velocity influence from previous panels)

$$A_{ij+1} = A_{ij+1} + \frac{x_{i+1}-x_i}{l_i} P_{22_{ji}} - \frac{y_{i+1}-y_i}{l_i} P_{12_{ji}}$$

This is the velocity influence contribution of node $j+1$ dotted in the normal direction (added to the current normal velocity influence from previous panels)

Let's populate part of one row! (Influence of all panels on first control point)

$$[\mathbf{P}] = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} = \begin{bmatrix} \text{influence of } \gamma_1 \text{ on } V_x & \text{influence of } \gamma_2 \text{ on } V_x \\ \text{influence of } \gamma_1 \text{ on } V_y & \text{influence of } \gamma_2 \text{ on } V_y \end{bmatrix}$$

$$\hat{n}_j = \frac{1}{\ell_j}\begin{Bmatrix} -t_y \\ t_x \end{Bmatrix} = \begin{Bmatrix} -\dfrac{y_{j+1} - y_j}{\ell_j} \\ \dfrac{x_{j+1} - x_j}{\ell_j} \end{Bmatrix} = \begin{Bmatrix} n_x \\ n_y \end{Bmatrix}$$

Control point $i = 1$

$A_1(\text{influence on cp } i) = [(\ \ )\quad (\ \ )\quad (\ \ )\quad (\ \ )\quad (\ \ )\quad (\ \ )]$

Panel $j = 1$

$\left(P_{11}n_x + P_{21}n_y\right)_{11}$

$\left(P_{12}n_x + P_{22}n_y\right)_{11}$

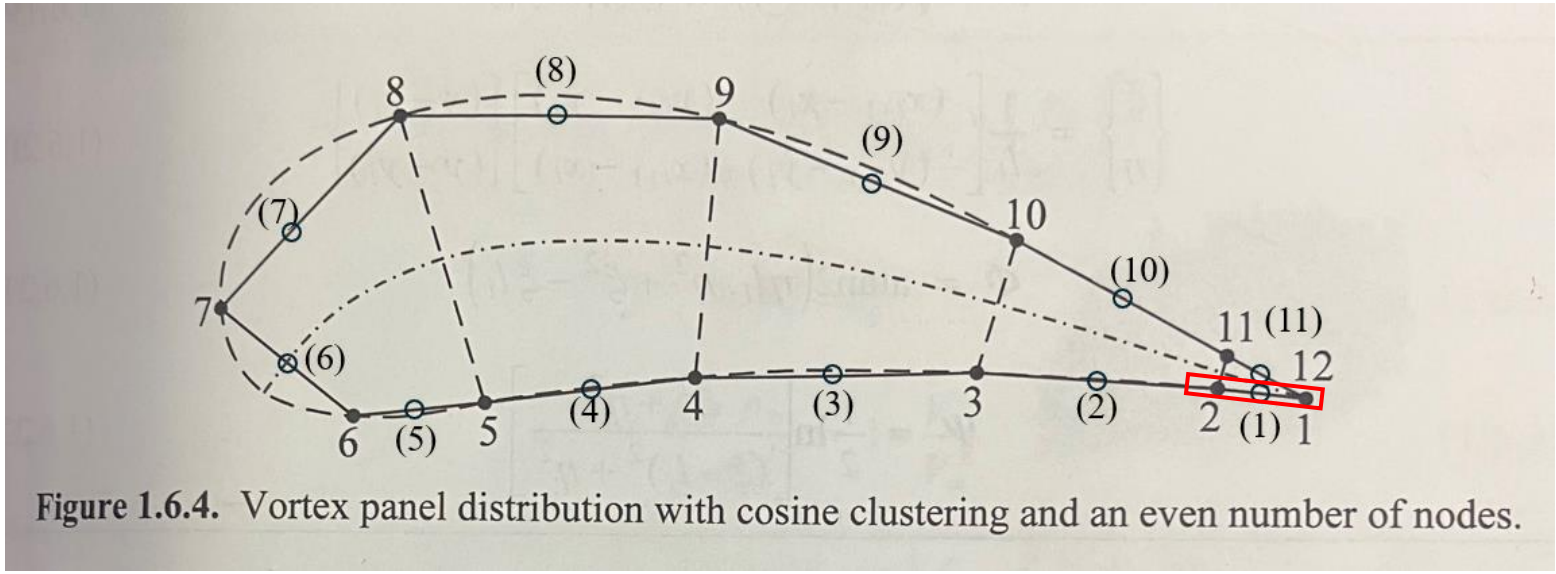


Figure 1.6.4. Vortex panel distribution with cosine clustering and an even number of nodes.

$$\left.\begin{aligned} A_{ij} &= A_{ij} + \frac{x_{i+1} - x_i}{l_i} P_{21_{ji}} - \frac{y_{i+1} - y_i}{l_i} P_{11_{ji}} \\ A_{ij+1} &= A_{ij+1} + \frac{x_{i+1} - x_i}{l_i} P_{22_{ji}} - \frac{y_{i+1} - y_i}{l_i} P_{12_{ji}} \end{aligned}\right\},$$

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$$\hat{n}_j = \frac{1}{\ell_j}\begin{Bmatrix} -t_y \\ t_x \end{Bmatrix} = \begin{Bmatrix} -\dfrac{y_{j+1} - y_j}{\ell_j} \\ \dfrac{x_{j+1} - x_j}{\ell_j} \end{Bmatrix} = \begin{Bmatrix} n_x \\ n_y \end{Bmatrix}$$

Control point $i = 1$

$A_1(\text{influence on cp } i) = [(\ \ )\quad(\ \ )\quad(\ \ )\quad(\ \ )\quad(\ \ )\quad(\ \ )]$

Panel $j = 1$

$\left(P_{11}n_x + P_{21}n_y\right)_{11}$

$\left(P_{12}n_x + P_{22}n_y\right)_{11}$

Panel $j = 2$

$\left(P_{11}n_x + P_{21}n_y\right)_{21}$

$\left(P_{12}n_x + P_{22}n_y\right)_{21}$

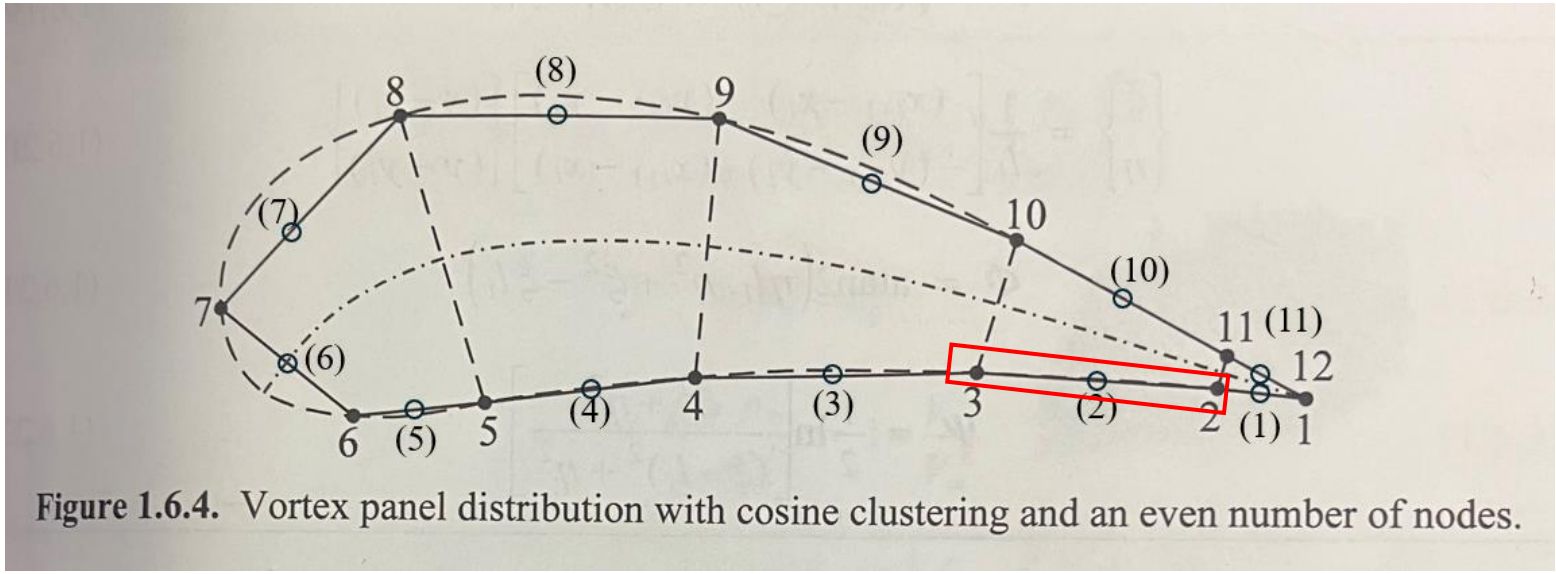


Figure 1.6.4. Vortex panel distribution with cosine clustering and an even number of nodes.

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Control point $i = 1$

$A_1(\text{influence on cp } i) = [(\ \ )\quad(\ \ )\quad(\ \ )\quad(\ \ )\quad(\ \ )\quad(\ \ )]$

Panel $j = 1$

$\left(P_{11}n_x + P_{21}n_y\right)_{11}$

$\left(P_{12}n_x + P_{22}n_y\right)_{11}$

Panel $j = 2$

$\left(P_{11}n_x + P_{21}n_y\right)_{21}$

$\left(P_{12}n_x + P_{22}n_y\right)_{21}$

Panel $j = 3$

$\left(P_{11}n_x + P_{21}n_y\right)_{31}$

$\left(P_{12}n_x + P_{22}n_y\right)_{31}$

Continues through all j!

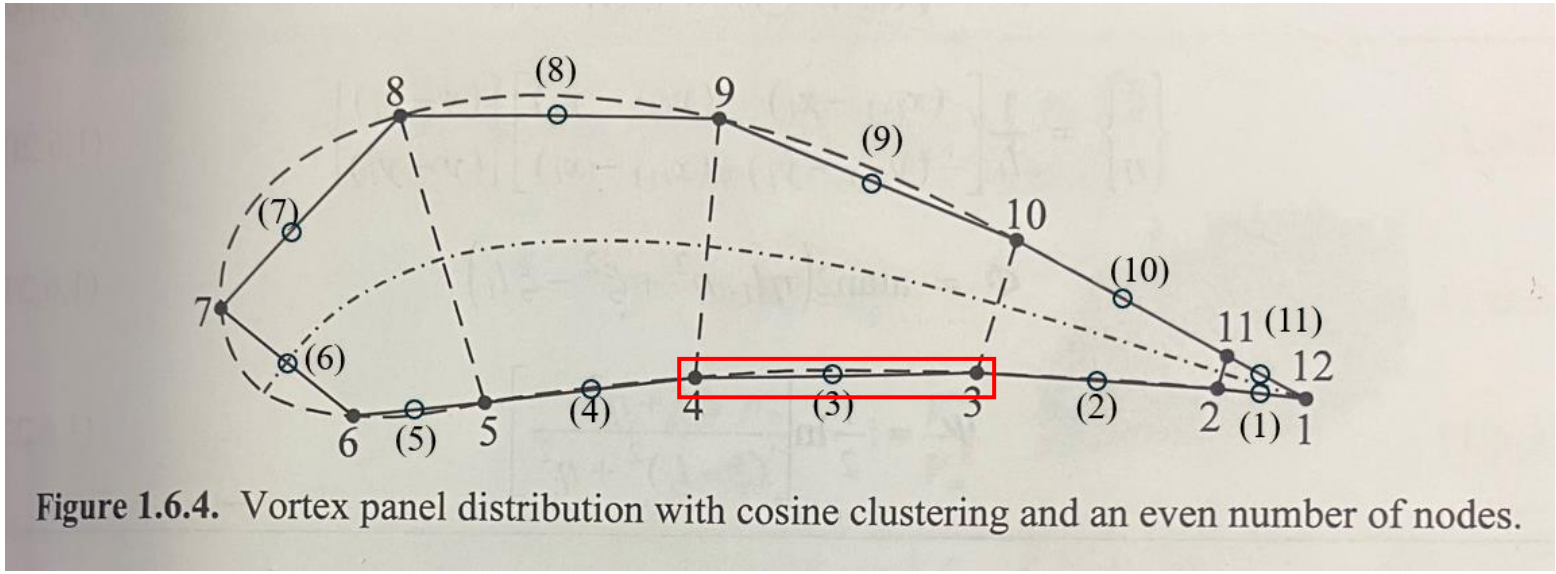


Figure 1.6.4. Vortex panel distribution with cosine clustering and an even number of nodes.

$$\left.\begin{aligned} A_{ij} &= A_{ij} + \frac{x_{i+1} - x_i}{l_i} P_{21_{ji}} - \frac{y_{i+1} - y_i}{l_i} P_{11_{ji}} \\ A_{ij+1} &= A_{ij+1} + \frac{x_{i+1} - x_i}{l_i} P_{22_{ji}} - \frac{y_{i+1} - y_i}{l_i} P_{12_{ji}} \end{aligned}\right\},$$